



MARIA-RAFAELLA ŞERB

Industrial Product Development Specialist and Doctoral Researcher in Industrial Engineering

National University of Science and Technology POLITEHNICA Bucharest – Piteşti University Centre

EDUCATION

PhD in Industrial Engineering (2025–present)

National University of Science and Technology POLITEHNICA Bucharest, Piteşti University Centre.

Doctoral research topic: *Contributions to the Development of an Intelligent Additive Manufacturing System for the Customization of Automotive Components in the Context of Industry 5.0*

MSc in Logistics Management (2023–2025)

Faculty of Mechanics and Technology, University of Piteşti. Dissertation: Optimising ergonomic conditions in the tyre manufacturing process.

BEng in Industrial Engineering and Management (2019–2023)

Faculty of Mechanics and Technology, University of Piteşti. Thesis: Design of the machining process for a bushing and a study on FIFO inventory management.

Technician in Economic Activities (2015–2019)

Maria Teiuleanu Economic College, Piteşti.

PROFESSIONAL EXPERIENCE

Product Industrialisation Specialist | Pirelli Tyres Romania, Slatina (March 2024–present)

Coordinates prototype tyre production and follows development from raw materials to final testing. Monitors quality, reviews test results, supports design adjustments and prepares technical documentation with research, testing and industrialisation teams.

Quality Specialist | Gestamp Beyçelik Romania, Piscani (December 2023–February 2024)

Monitored product quality and blocked products; participated in internal and external audits; prepared work instructions, processed SAP transactions and documented capability assessments based on 3D measurements.

Quality Assurance Technician | Bavasi Plast Group, Mioveni (March–September 2023)

Performed routine inspection, consolidated quality control records, conducted 5S, product and process audits, and prepared workplace control and packaging documentation.

Quality Department Intern | Leoni Wiring Systems, Piteşti (September 2022–March 2023)

Supported document control, wiring harness measurements, quality control records and Excel data entry.

RESEARCH INTERESTS

Industrial ergonomics; additive manufacturing; human-machine collaboration; product industrialisation; quality management; logistics and process improvement.

PUBLICATIONS

2025 Șerb, M.-R.; Anghel, D.-C. Integration of Artificial Intelligence and Human-Machine Collaboration in Additive Manufacturing: A Review in the Context of Industry 5.0. Scientific Bulletin – Automotive Series, 35. <https://doi.org/10.26825/bup.ar.2025.005>

2024 Șerb, M.-R.; Sicoe, G.-M.; Anghel, D.-C. Advanced Perspectives on Ergonomic Risk Assessment: From Classical Techniques to Intelligent Systems. Scientific Bulletin – Automotive Series, 34. <https://doi.org/10.26825/bup.ar.2024.007>

SCIENTIFIC AND PROFESSIONAL ACTIVITIES

2025 – Student Scientific Communications Session, POLITEHNICA Bucharest, Pitești University Centre: Ergonomic optimisation of tyre manufacturing conditions.

2023 – Student Scientific Communications Session, University of Pitești: Implementing FIFO for inventory management.

2023 – Problem Solving Workshop / Hackathon, Arctic S.A.

TECHNICAL SKILLS

Quality audits (5S, product and process); quality control documentation; SAP; CATIA V5 and AutoCAD (intermediate); Microsoft Office (intermediate); Outlook, Teams and Zoom.

LANGUAGES

Romanian: native | English: B1 | French: A2

Date: September 2026